Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df048-jun-android
Operator:	Floris
Analysis Started:	2026-06-15 15:01:41
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\dfdof_output\CASE-df048-jun-android
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	395.9 MB	2026-06-15 15:01:43
df048_external.E01	drone_sd	physical	7.8 GB	2026-06-15 15:02:06
df048_internal.001	drone_sd	physical	8.0 GB	2026-06-15 15:02:38

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\android

Cryptographic Hashes

android_logical.zip	controller_android	logical
SHA-256:	9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d	
SHA-1:	2b2e6a7c2d51ea06c77e2c0015052f3ef0335f5d	

df048_external.E01	drone_sd	physical
SHA-256:	435b97c0cba3007d40b4ac10686049d1527fa0dc1235d8492a2df3e66d8bdac9	
SHA-1:	f46a295ef957df0dc3616ea5a95567f8df33e05b	

df048_internal.001	drone_sd	physical
SHA-256:	71b288d9b8e639a9b9f7c788dfe6bf71a148e8a4c93377b59cb36c35113a8ca6	
SHA-1:	8ae48fb181c3cf3c78e2e9680366f625d860a4a0	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_external.E01; treating as direct partition image
2. [p1 - drone sd]: mmls: no partition table found in df048_internal.001; treating as direct partition image
3. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
4. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
5. [p2 - controller ios]: source evidence not found
6. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
7. [p3 - controller android - account data]: no artefacts found
8. [p3 - controller android - databases]: no artefacts found
9. [p3 - controller ios]: source evidence not found
10. [p3 - drone flight storage]: source evidence not found
11. [p4 - controller ios]: source evidence not found
12. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	2	2	[OK]
exiftool	28	28	[OK]
fls	2	2	[OK]
icat	20	20	[OK]
mmls	2	0	[X] Failed
parser_android	1	1	[OK]
txtlogtocsvtool	2	2	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	1 / 3
Tools Succeeded	6 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Drone Serial Number	0670136230 0K1DF313BD4PR1	[!] Inconsistent	2
Drone Model	Matrice 600 Pro Mavic Air MavicAir	[!] Inconsistent	4
DJI Application Version	3.1.11 4.2.16	[!] Inconsistent	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: eb147a2263a033419d84a1ddf76ec224bb14d75afd3d7e60721913917a4be493
Flight Record Name: DJIFlightRecord_2017-10-10_[10-02-22].csv
Start: 2017-10-10T16:02:22.901000+00:00
End: 2017-10-10T16:07:59.866000+00:00
Duration: 337.0 s

Flight Metrics

Peak Height: 35.3 m
Distance Travelled: 2723.32 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-10-10 16:02:22	ctrl_and:fr (v3.1.11)	Log started	SN: 0670136230; App: 3.1.11; Matrice 600 Pro
2017-10-10 16:02:22	ctrl_and:fr (v3.1.11)	Motor turned on	on
2017-10-10 16:02:23	ctrl_and:fr (v3.1.11)	Record mode changed	No
2017-10-10 16:02:47	ctrl_and:fr (v3.1.11)	Record mode changed	Starting
2017-10-10 16:03:48	ctrl_and:fr (v3.1.11)	Reached peak height	35.3 m
2017-10-10 16:07:59	ctrl_and:fr (v3.1.11)	Log ended	Dist home: 2.0 m; Height: -1.5 m; Rec: 0.0 s

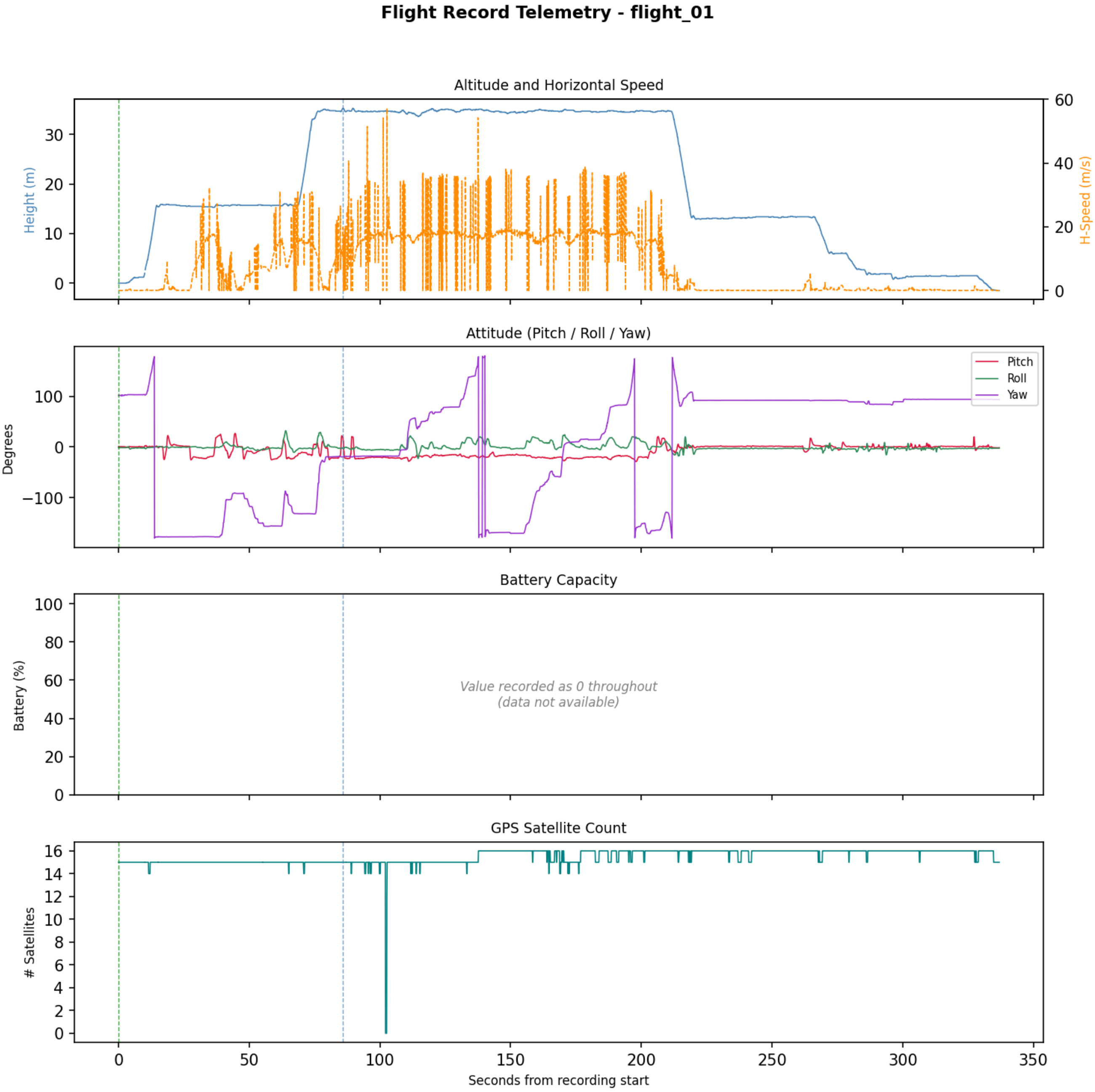
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	

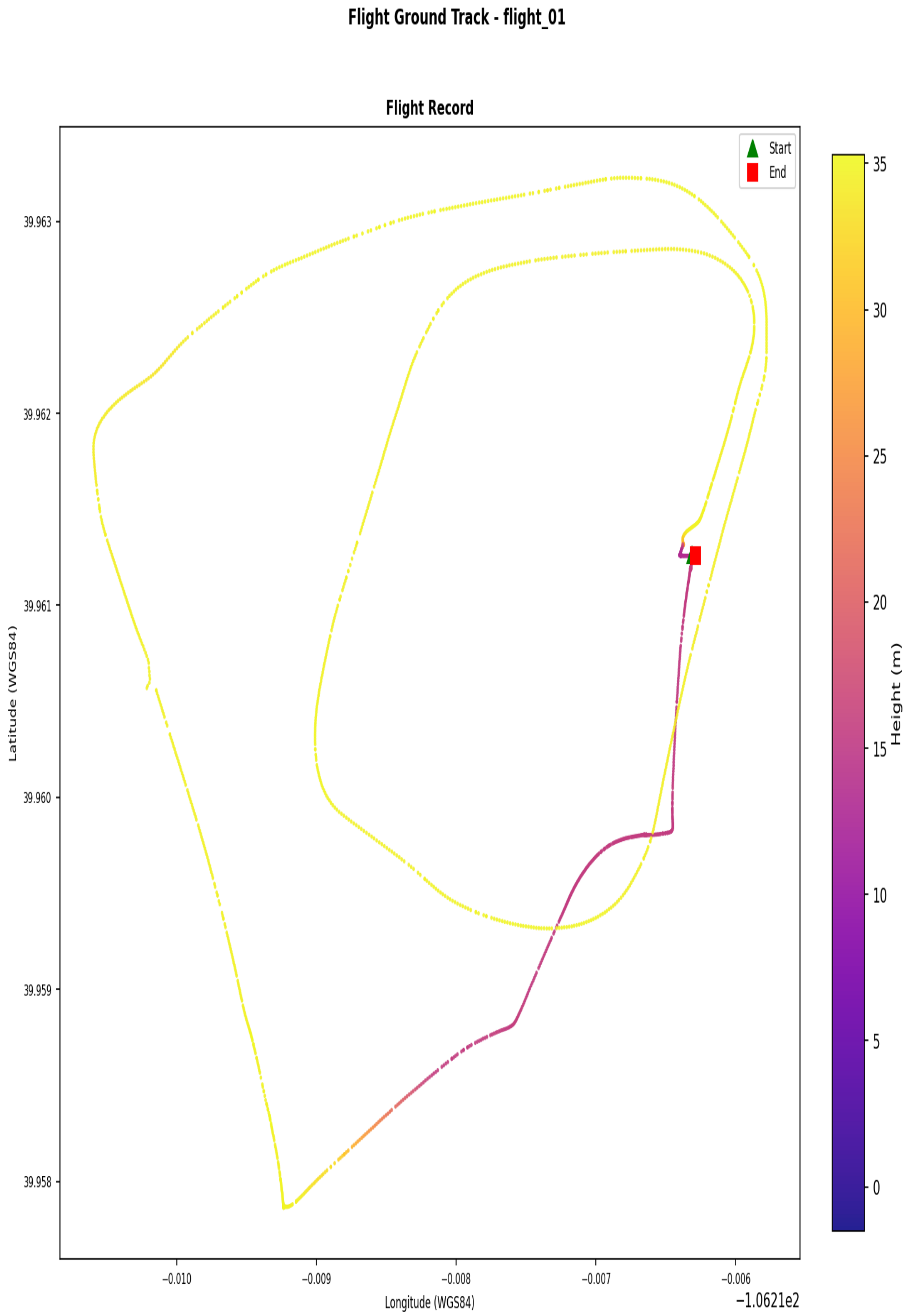
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	

Flight Record - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	305.4 s
Median GPS Distance:	1.75 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	`f36d1f9b2e295eccb028cfca3a19b6a8271452d872c73494631ce0dc7bed8845`
Flight Record Name:	DJIFlightRecord_2018-06-19_[14-50-34].csv
Start:	2018-06-19T18:50:34.869000+00:00
End:	2018-06-19T18:55:40.243000+00:00
Duration:	305.4 s
Drone Log SHA-256:	`9a760be61df4a375716f0485b44c3d4cb6a0fe21c6c6abc591f76768fdbfa434`
Drone Log Name:	18-06-19-02-47-38_FLY057.csv
Start:	2018-06-19T18:46:00.048000+00:00
End:	2018-06-19T18:58:21.725000+00:00
Duration:	741.7 s

Flight Metrics

Peak Height:	32.196024004978064 m
Distance Travelled:	444.56 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	300.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. ASST_TAKEOFF
3. EngineStart
4. AutoTakeoff
5. AutoLanding
6. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_and:dl	Motor turned off	off
2018-06-19 18:49:18	ctrl_and:dl	Log started	MavicAir
2018-06-19 18:50:34	ctrl_and:dl	Motor turned on	on
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Log started	SN: 0K1DF313BD4PR1; App: 4.2.16; Mavic Air
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Motor turned on	on
2018-06-19 18:50:34	ctrl_and:fr (v4.2.16)	Record mode changed	No
2018-06-19 18:50:39	ctrl_and:fr (v4.2.16)	Record mode changed	Starting
2018-06-19 18:54:25	ctrl_and:dl	Reached peak height	32.2 m
2018-06-19 18:54:26	ctrl_and:fr (v4.2.16)	Reached peak height	28.5 m
2018-06-19 18:55:39	ctrl_and:dl	Motor turned off	off
2018-06-19 18:55:40	ctrl_and:fr (v4.2.16)	Log ended	Dist home: 0.7 m; Height: 0.0 m; Rec: 300.0 s
2018-06-19 18:55:40	ctrl_and:fr (v4.2.16)	Motor turned off	off
2018-06-19 18:58:23	ctrl_and:dl	Log ended	

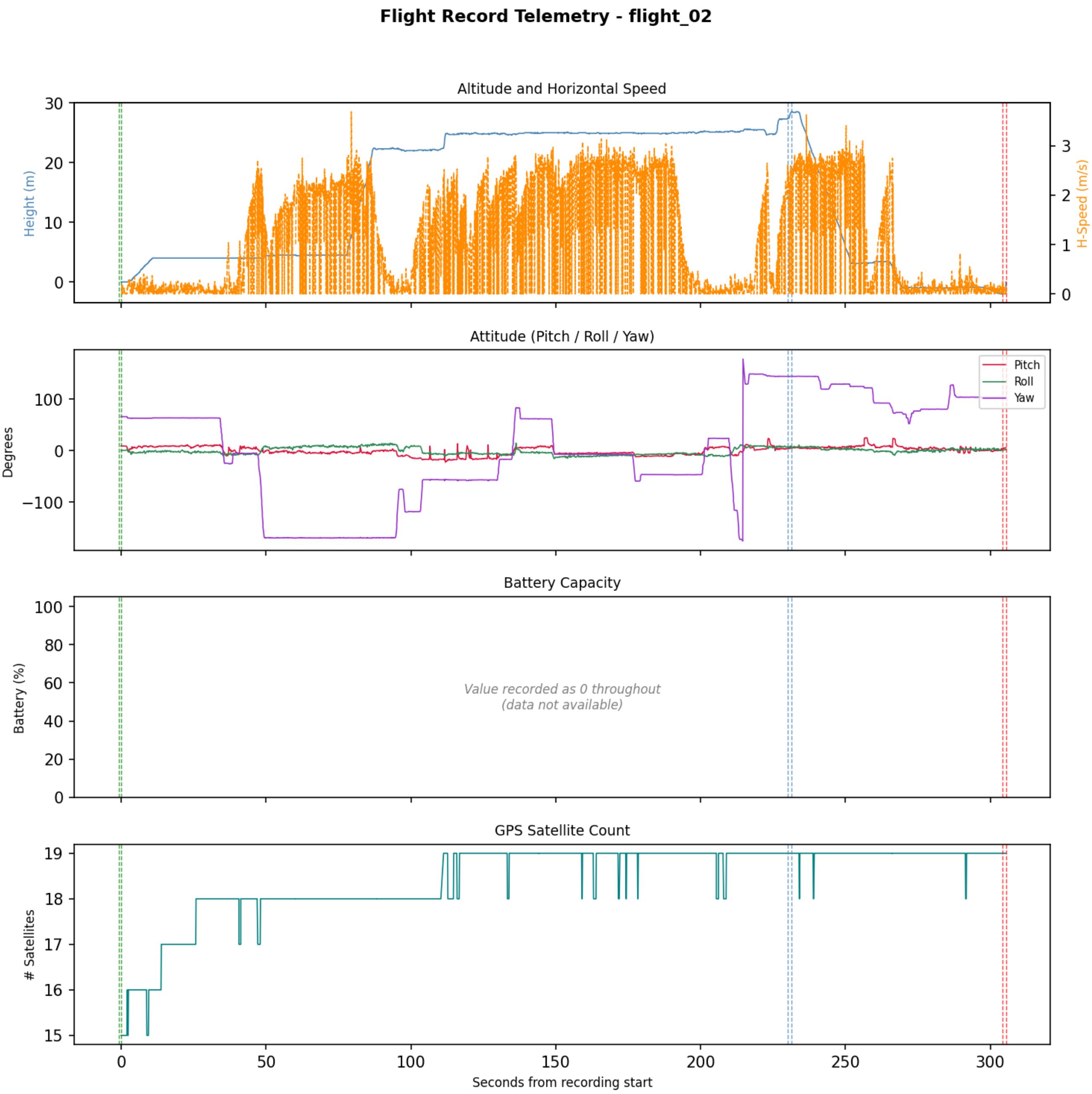
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	

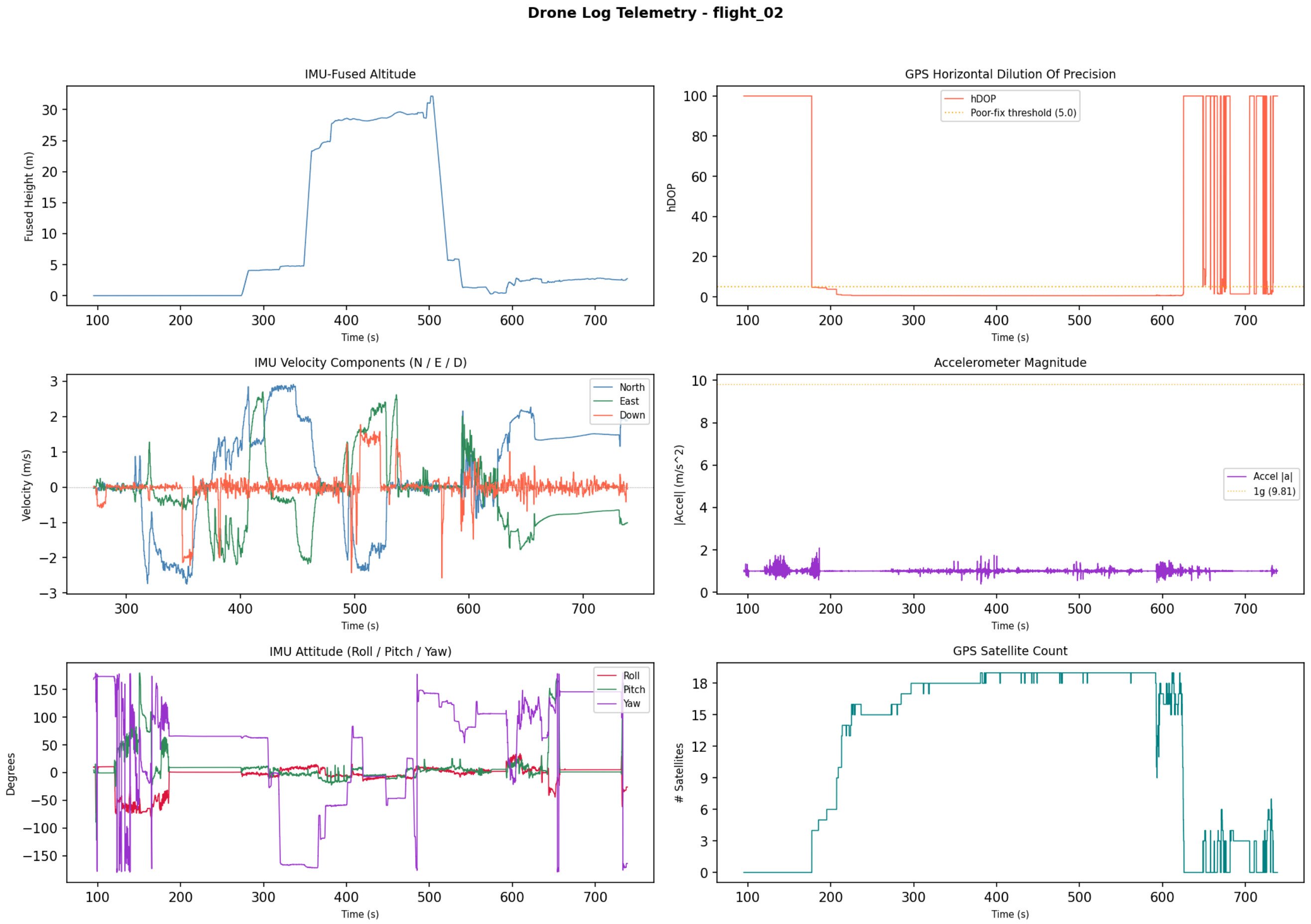
controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	

Flight Record - Sensor Telemetry



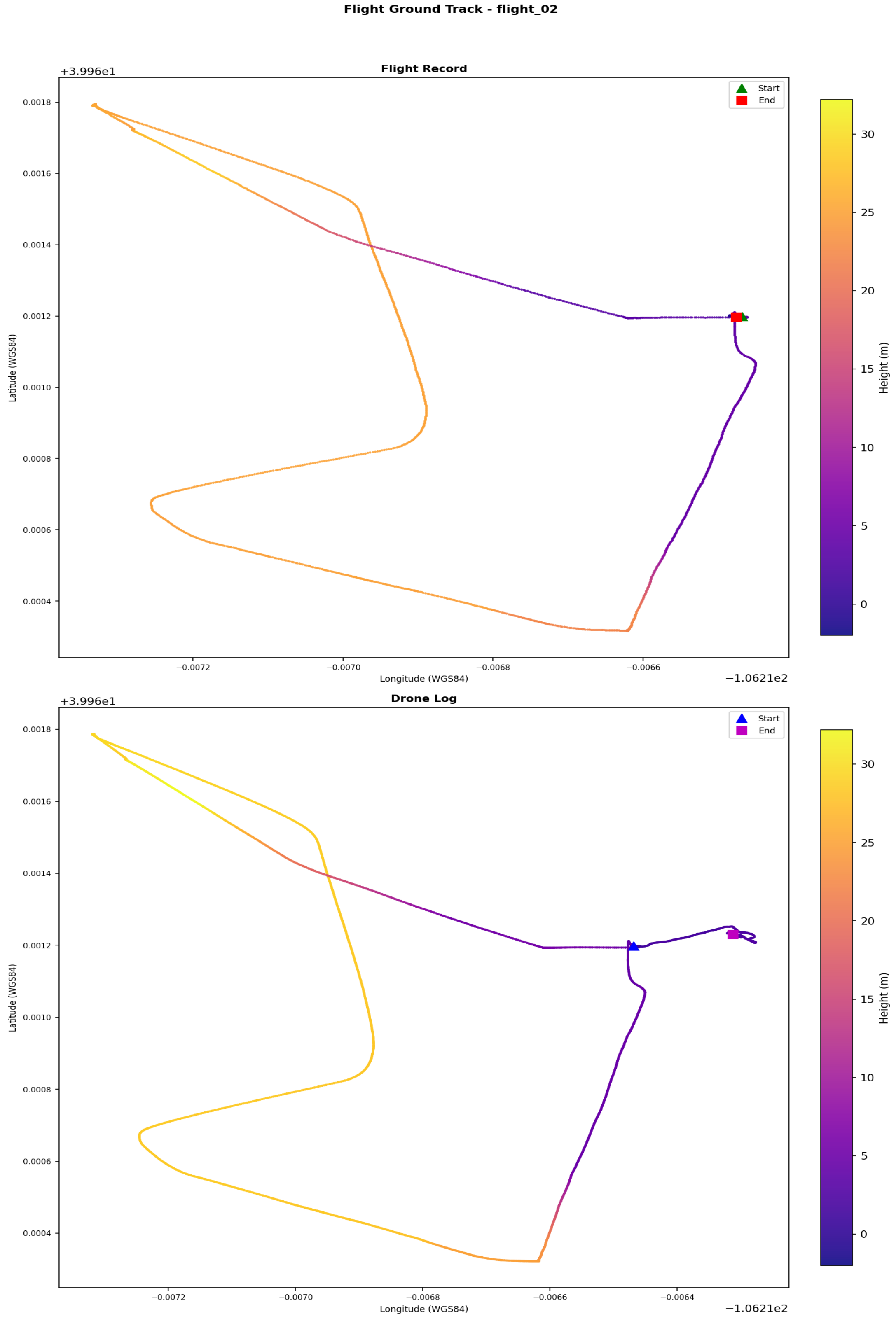
GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 18-06-19-02-47-38_FLY057.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (18-06-19-02-47-38_FLY057.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`ffec66cca6b5833e946d72ef1f95195c7e8ed2aafb4c0bc48ee3e8e5efcf9571`
Drone Log Name:	18-06-19-02-59-05_FLY058.csv
Start:	2018-06-19T18:58:38.544000+00:00
End:	2018-06-19T19:02:09.813000+00:00
Duration:	211.3 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

Fly Modes Observed (Chronological)

1. GPS_Atti
2. Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_and:dl	Reached peak height	-0.0 m
-	ctrl_and:dl	Motor turned off	off
2018-06-19 19:00:23	ctrl_and:dl	Log started	MavicAir
2018-06-19 19:02:08	ctrl_and:dl	Log ended	

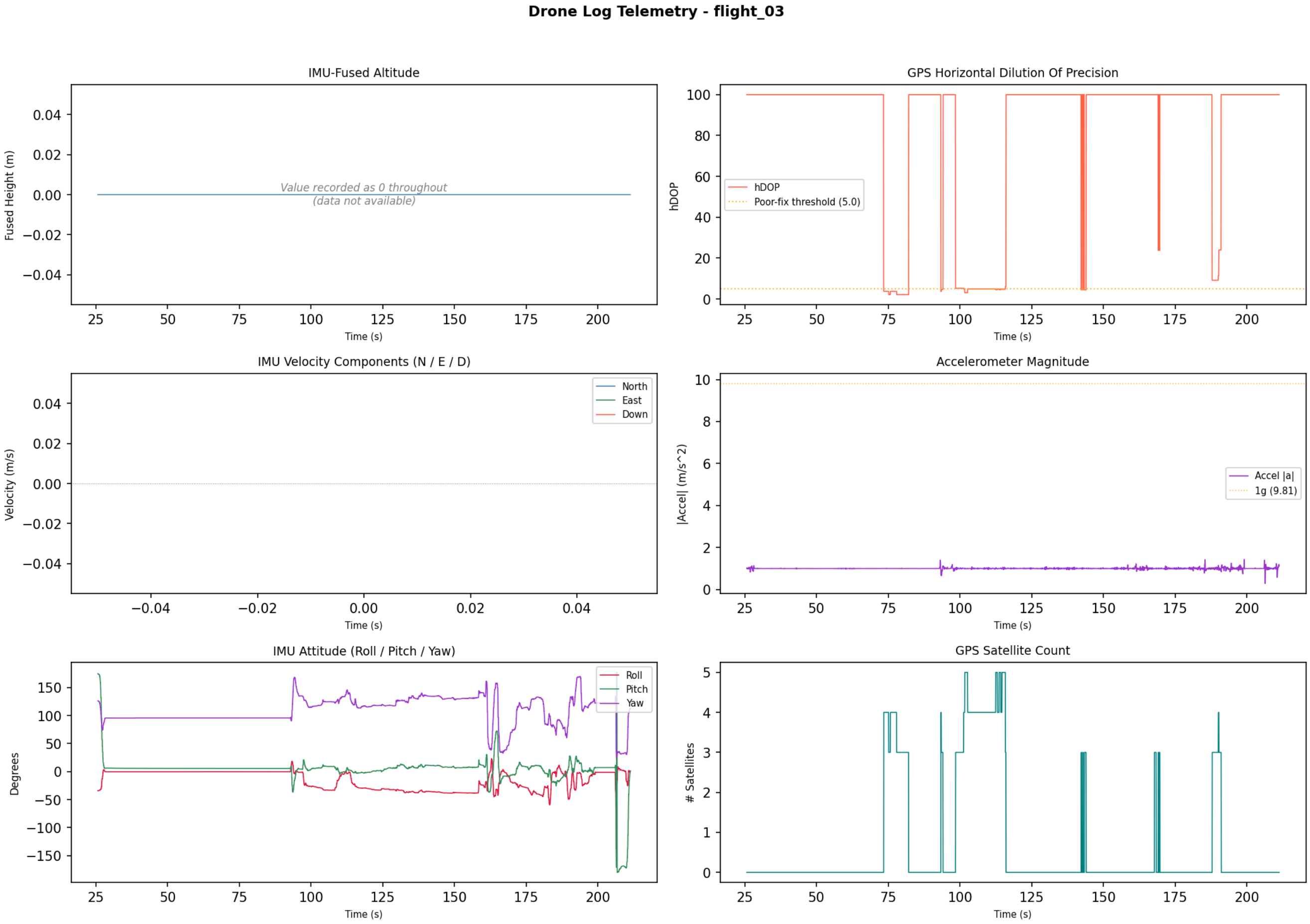
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2163)
p5:0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345	

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	0	0	2	33	2	6	4
drone_sd	0	0	0	0	0	10	10

Data Quality Notes per Category

Drone Logs (2 files):

- 2 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Flight Logs (33 files):

- 1 file(s) unreadable (binary)
- formats: 2 heading_log, 30 unknown

Flight Records (2 files):

- 2 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Images (16 files):

- 16 file(s) missing (normalisable) EXIF timestamp
- 16 file(s) missing EXIF GPS coordinates

Videos (14 files):

- 2 file(s) contain(s) EXIF GPS coordinates
- 12 file(s) missing (normalisable) EXIF timestamp
- 12 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 2 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 2 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 2 heading_log, 30 unknown.
- Flight_records: 2 file(s) with row anomalies detected in telemetry data.
- Flight_records: 2 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_external.E01; treating as direct partition image
2. [p1 - drone sd]: mmls: no partition table found in df048_internal.001; treating as direct partition image
3. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
4. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
5. [p2 - controller ios]: source evidence not found
6. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
7. [p3 - controller android - account data]: no artefacts found
8. [p3 - controller android - databases]: no artefacts found
9. [p3 - controller ios]: source evidence not found
10. [p3 - drone flight storage]: source evidence not found
11. [p4 - controller ios]: source evidence not found
12. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Drone Logs

norm_18-06-19-02-47-38_FLY057.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 4 row(s) affected
- motor_airborne_off: 2614 row(s) affected
- gps_poor_fix: 3327 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATTI(0):absoluteHeight:C, IMU_ATTI(0):distanceHP:C, IMU_ATTI(1):absoluteHeight:C (+1 more)
- contains_constant_value: 39 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:BatAuth:D, BatteryInfo:FullCap:D, BatteryInfo:LowVolThreshold:D (+34 more)
- gps_accuracy_poor: 2790 row(s) affected
- attitude_out_of_bounds: 513 row(s) affected

norm_18-06-19-02-59-05_FLY058.csv

- missing_gps: 4 row(s) affected
- gps_poor_fix: 3400 row(s) affected
- contains_no_value: 28 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+23 more)
- contains_constant_value: 90 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:comp_alti:D, AirSpeed:vel_level:D (+85 more)
- gps_accuracy_poor: 3276 row(s) affected
- attitude_out_of_bounds: 119 row(s) affected

Flight Logs

10-10-2017

- format: heading_log | entries: 7

19-06-2018-0K1DF313BD4PR1
- format: heading_log | entries: 7

log-2017-09-27.txt
- format: unknown | entries: 0

log-2017-09-27_1.txt
- format: unknown | entries: 0

log-2017-09-27_2.txt
- format: unknown | entries: 0

log-2017-09-27_3.txt
- format: unknown | entries: 0

log-2017-09-27_4.txt
- format: unknown | entries: 0

log-2017-09-27_5.txt
- format: unknown | entries: 0

log-2017-09-27_6.txt
- format: unknown | entries: 0

log-2017-09-27_7.txt
- format: unknown | entries: 0

log-2017-10-09.txt
- format: unknown | entries: 0

log-2017-10-09_1.txt
- format: unknown | entries: 0

log-2017-10-09_2.txt
- format: unknown | entries: 0

log-2017-10-09_3.txt
- format: unknown | entries: 0

log-2017-10-09_4.txt
- format: unknown | entries: 0

log-2017-10-09_5.txt
- format: unknown | entries: 0

log-2017-10-09_6.txt
- format: unknown | entries: 0

log-2017-10-09_7.txt
- format: unknown | entries: 0

log-2017-10-10.txt
- format: unknown | entries: 0

log-2017-10-10_2.txt
- format: unknown | entries: 0

log-2017-10-10_3.txt
- format: unknown | entries: 0

log-2017-10-10_4.txt
- format: unknown | entries: 0

log-2017-10-10_5.txt
- format: unknown | entries: 0

log-2017-10-10_6.txt
- format: unknown | entries: 0

log-2017-10-10_7.txt
- format: unknown | entries: 0

log-2017-10-10_8.txt
- format: unknown | entries: 0

log-2017-12-11.txt

- format: unknown | entries: 0

log-2017-12-11_1.txt

- format: unknown | entries: 0

log-2017-12-11_2.txt

- format: unknown | entries: 0

log-2017-12-12.txt

- format: unknown | entries: 0

log-2017-12-12_1.txt

- format: unknown | entries: 0

log-2017-12-12_2.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-10-10_[10-02-22].csv

- altitude_negative: 49 row(s) affected
- gps_poor_fix: 5 row(s) affected
- contains_no_value: 35 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh], CENTER_BATTERY.currentPV [V] (+30 more)
- contains_constant_value: 167 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+162 more)

norm_DJIFlightRecord_2018-06-19_[14-50-34].csv

- altitude_negative: 367 row(s) affected
- contains_no_value: 58 column(s) - APP_SER_WARN.warn, APP_TIP.tip, CENTER_BATTERY.connStatus, CENTER_BATTERY.current [A], CENTER_BATTERY.currentCapacity [mAh] (+53 more)
- contains_constant_value: 156 column(s) - APP_GPS.longitude, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+151 more)

Images

thumb_33177fb665068406_1523611394000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_5be64acbef28816c_1523610990000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_6d9f39f0ca9decc3_1523610988000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_7f92b744bbc53b3a_1523610992000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_dd609e61934788b3_1523611494000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_ed8c1a297e1061b4_1524152656000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2017_10_10_10_02_47.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2018_06_19_14_50_39.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Drone Serial Number inconsistency detected - conflicting values observed: 0670136230, 0K1DF313BD4PR1. Manual verification required.
4. Drone Model inconsistency detected - conflicting values observed: Matrice 600 Pro, Mavic Air, MavicAir. Manual verification required.
5. Note: no controller device found - Device and account information.
6. DJI Application Version inconsistency detected - conflicting values observed: 3.1.11, 4.2.16. Manual verification required.
7. Note: no account e-mail found - Device and account information.
8. Note: no product type found - Device and account information.
9. Note: no firmware version found - Device and account information.
10. Note: no last country code found - Device and account information.
11. Note: no aircraft model code found - Device and account information.
12. Note: no controller serial number found - Device and account information.
13. Note: no last known latitude found - Device and account information.
14. Note: no last known longitude found - Device and account information.
15. Note: no account uid found - Device and account information.
16. Note: no account nickname found - Device and account information.
17. Note: no device uuid found - Device and account information.
18. 2 flight record(s) could not be correlated to a drone log (flight_01, flight_03). Independent corroboration is absent for these flight(s).

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

18-06-19-02-47-38_FLY057.DAT	Drone Logs	controller_android	extract_logical
SHA-256:	06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1		
SHA-1:	6f587093e526ec9f283190c1d33b04a0c206d470		

18-06-19-02-59-05_FLY058.DAT	Drone Logs	controller_android	extract_logical
SHA-256:	fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a		
SHA-1:	249331ad93c487a35571f531c4448907ce4e2ddd		

19-06-2018-0K1DF313BD4PR1	Flight Logs	controller_android	extract_logical
SHA-256:	cb9671659ae2ff11b0fd3dd846e8702a81366479b04f3e2442300b2211bcff3e		
SHA-1:	3792e2ff77e3cecca7dd66823748139f459bd108		

log-2017-09-27.txt	Flight Logs	controller_android	extract_logical
SHA-256:	894f25c259709491d0a768c24bd0fc99ed9f4793d25d17c006024683817792bf		
SHA-1:	f15b75a35193a648e2a8c9a625aa8e45077f174e		

log-2017-10-09.txt	Flight Logs	controller_android	extract_logical
SHA-256:	4961f9bd65d1da4df06918bb4aaa7933feb7c3c0189f03e7bbf477c9160c3750		
SHA-1:	2c678591d4c3d6f98c1e671dd0b95a9580151132		

log-2017-10-10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	87e7baf7d274fff12e46e1de5db451f450cee31655b22f6c07c34591742774bc		
SHA-1:	082875e1cfd4f72f0a2b8bb40ca1c2accc868372		

log-2017-09-27_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9d2a843316232069c58ec9b794369f2d1e174ed5a9a681df9c3e92a63e21aaa9		
SHA-1:	7f860f235699bf30368d06735c86c4770163a883		

log-2017-10-09_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	e36c5fa294f38c8e5ce12322e922365150ee5fbae22ac26db11781bdb3de94c8		
SHA-1:	755e6ee81cecb9cd2e51f01e32d3e3b3a73fa768		

log-2017-10-10_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	ae84503564a90d25c78355de393d7c7a7efa6d7731ddc9b7f41cabca39953187		
SHA-1:	3d50cf703e7e39252101d91a7c74cf587c853134		

log-2017-12-11.txt	Flight Logs	controller_android	extract_logical
SHA-256:	96f5a059fb10069371ed62066f00ca0661d1fc5a02c170a48a8cceb7d7447604		
SHA-1:	9a2d18560b053ddbf03180a05f7715f0122483d1		

log-2017-12-12.txt	Flight Logs	controller_android	extract_logical
SHA-256:	bbe0eda5d1f58d893d5bdbb1355df2dcd5f0607e83c7a012e0b19ebe39999b7f		
SHA-1:	9e73a56dded6994d8bab87c6e0b36fb0c28fcc8d		

log-2017-09-27_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	2baea06c4aebba79a3d66fc9ecc1443785b551d9c7bf8ec01a764eb1cdeb501a		
SHA-1:	b5dc41909888ac0a72fbbb75389945e82d73d2b7		

log-2017-10-09_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c8341a3abc580a827cfff3e9713e904b8d781a4a930007b67e61f5803b60e63e		
SHA-1:	b9c918529b0fcc1968a1edaa3f14ccfe855e671a		

log-2017-10-10_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	781f12c3c4e32cec55f04d45f6d1ea7cce95d23049f4ed66329cca98ea419270		
SHA-1:	a7f735c63f556d42ec358564daa0aaaf49c19df0		

log-2017-12-11_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	8f97e22370768db84494a0db003fad1056d5a29e747f6c49dafa20ee6b594926		
SHA-1:	4ecab73d120e2fb79ec11c082a8ca3bd23dd8f0a		

log-2017-12-12_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	ab4d29e99e718c141f9c9f85d211dc1827169f3fa6e65d749e4922c17c6be004		
SHA-1:	e7fdd9f6b2f537fb983df3316e23e321febd099d		

log-2017-09-27_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	7a60de6d0104d8d1673a7e57f1e6d0f128785e032076426563ec469ebba23a39		
SHA-1:	382f017cb1d7ba79ef6457b288fb6ff5f9519652		

log-2017-10-09_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	7724ba97cd4511eb2b0843aa8efdc5bd7192d2ff9c2fd3c5f3a4dbb3d81ada8f		
SHA-1:	e8745e71fe2a63c3f3817ffe18028dd672a0f0bf		

log-2017-10-10_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	3a70004e5bccd9e5dd80f8d96d53dbab7bfdf31f3338f81964742584cb77fc56		
SHA-1:	ff114795961a33488379f9a7c4144503d899df19		

log-2017-09-27_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	d90e302e33743d784de0ef46a13a17beb71fd40b217e632195f1d1b19bfb870c		
SHA-1:	2774c8e36266f759189106d5f8ef05a9643022be		

log-2017-10-09_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	bc0ee9e645bba2510f45462f13cdb2973e48c0abb880c79cfd53862229f7a633		
SHA-1:	1f83e5e1e0a0aba13153a410c74ab6ccad1dfc2f		

log-2017-10-10_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	f37f26fa51ad3027d858170b91eb4f5daf5dd824dcedb994e4f2d0b8dec8f474		
SHA-1:	8565ef7f3cede81d6102aa05193c15a900c0832f		

log-2017-12-11_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	8bd579a33bd9246a9452fe243f851292d8b178607d625cd8946082d3aaf229a1		
SHA-1:	88b4283d8696d180cae205760f144935203b5cb9		

log-2017-12-12_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	87ed69dea6386ec3cc6b6cd7059bc7afe8444a6db49b91fab9fb23defba93c24		
SHA-1:	fc465cc7d2098e1532d58ac43d747785e2a2d424		

log-2017-09-27_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	0038111eaa570a867f06b0d10f1fa2dff3c3ac967c5af10941aa0423526dd843		
SHA-1:	7488b69ac447860fc1d844d81fd4e42de4b84611		

log-2017-10-09_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	d458d39b5a7a3c3cd2b3d277dd3bb1f6d628bf566bbe8daf26e98857485b4d71		
SHA-1:	210d44431515e4f8b6d3dd6648ab5591ca575a3f		

log-2017-10-10_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c26f9650e93c2733f7316978f2cd3a370b2aedbc3db86bc4821fe6087cb487bc		
SHA-1:	d6dd690bc097b55ecad370a9f201a0cfd7e3d1dd		

log-2017-10-10_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9e8e422fb7542b2d62fec5478852da23ecddf40dd08d177806da46cf3b619fac		
SHA-1:	4c68bf837ead4a5de9fd913b52cb3ad9b8559a2f		

log-2017-09-27_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	95fcab9567034a565b6bed21a8169e645846f97c8c033127d2a835733f9e78fc		
SHA-1:	10690bcf6ca5e428c11e1e10876cb70295a8b2aa		

log-2017-10-09_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	2a24f2bd2096630c94171eaff4ee623d6dd060697250e81124b6bae7874fbd23		
SHA-1:	a398ba235179b487991c850d811463ce91a8569e		

log-2017-10-10_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9876a533ddde0c5ad25489905bd3e0e35bce9a4d0f1f77aaebec2eaa9fb287ae		
SHA-1:	a44441dfa2f1e437087bc0d1fc6adf31d1ff89c7		

log-2017-09-27_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	662ea13bf63cc9787d4d843305e204a3c6734350fcae3368f7edb1f17e17f585		
SHA-1:	202fb4e595461cdf82c1aa0da24abca3a45e4f3d		

log-2017-10-09_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c578617197a4f0ff7f52391db1c29ac456712a3f5938f6315a8a4de174df353e		
SHA-1:	80e4508e292f6a494fb38d23182d7615b0b01eb0		

log-2017-10-10_8.txt	Flight Logs	controller_android	extract_logical
SHA-256:	4453dd1014b67ec7f35298f67ec559d96fb29e40aa8d5a16fa346469d011749a		
SHA-1:	9bdb40c1cf52b996ee1bdcf3230126cd3739f738		

10-10-2017	Flight Logs	controller_android	extract_logical
SHA-256:	48a35a1d275e0db3581766f72e8179401a4f74ea1bca7815942c659d86aea50a		
SHA-1:	a4d426fbf5f10d0b4375c5b4a082eb0819eaf4a0		

DJIFlightRecord_2018-06-19_[14-50-34].txt	Flight Records	controller_android	extract_logical
SHA-256:	0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a		
SHA-1:	66bc3fce709dff20df9fabcaa97a036d2c8f2c57		

DJIFlightRecord_2017-10-10_[10-02-22].txt	Flight Records	controller_android	extract_logical
SHA-256:	f134bef34e6877ae5a030297f56ae6e12fe946b468c5143ef9976c08170295f8		
SHA-1:	53b8c7d8257402a8df7649c5549c70252b726138		

thumb_33177fb665068406_1523611394000.jpg	Images	controller_android	extract_logical
SHA-256:	e8d6ac3d4cda2bccff4dd3e64950e87fd68700cf80b37f2c45e9e0ac59a4d42a		
SHA-1:	55b6683fe7bf253b609ca32c4fcc3a33795ca21d		

thumb_5be64acbef28816c_1523610990000.jpg	Images	controller_android	extract_logical
SHA-256:	e6a060c7756c7e2fd99c850434b02c708a329ebf16a814e0fe0a77b15caaab24		
SHA-1:	ed1bfdceab3e02d87575cadd6c79fdd983021271		

thumb_6d9f39f0ca9decc3_1523610988000.jpg	Images	controller_android	extract_logical
SHA-256:	cbe6209aa64bbb44340e4aed5d2e276fd1df75805f956307b24a83e79b1d6f90		
SHA-1:	5b90ca57716f15045a6a65c7215ca77036ff6af2		

thumb_7f92b744bbc53b3a_1523610992000.jp	Images	controller_android	extract_logical
SHA-256:	32ff09733d14f4ecdd59171811fd50111cc00752ebdb528f04b75b08cf220874		
SHA-1:	874dff84aaa18baf06b4070276135346c328e178		

thumb_dd609e61934788b3_1523611494000.jp	Images	controller_android	extract_logical
SHA-256:	8986d1b4a162ca4bbe0adbf12f01141b9d7673ba196436ee31e16fb6b540b2e2		
SHA-1:	3802402288db8466063ecaec81012031de544392		

thumb_ed8c1a297e1061b4_1524152656000.jp	Images	controller_android	extract_logical
SHA-256:	924f798a8f815a2a8c3862bf40481f34194ce30fbd637c939e4aa66fcb108fd3		
SHA-1:	898544f8b343f3ad616898273468c7203082872e		

2018_06_19_14_50_39.info	Videos	controller_android	extract_logical
SHA-256:	a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a		
SHA-1:	def346de1f2d68bb135f621930cfec9eca413581		

2018_06_19_14_50_39.mp4	Videos	controller_android	extract_logical
SHA-256:	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00		
SHA-1:	8afc6c44c37db8cb428875a43710b45f9549479e		

2017_10_10_10_02_47.info	Videos	controller_android	extract_logical
SHA-256:	0b7295423d1f0ac14065b46ef4e68d28afc12a858b37531fe545b4c20b8a9345		
SHA-1:	4ed6541a3da7544dded2162f25785a5a4b9a90d6		

2017_10_10_10_02_47.mp4	Videos	controller_android	extract_logical
SHA-256:	ce078b575399110fb022fac5c4802bf50dfe0f58e76976480c93aefee354e5dd		
SHA-1:	8be2153aba2e3419c8296774f6a64e61e52293c0		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0		
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f		
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79		
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4		
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f		
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815		
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea		

DJI_0002.MP4	Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b		
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f		

DJI_0003.MP4	Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a		
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beaacb3		
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192		
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0		
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f		
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79		
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4		
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f		
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815		
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea		

DJI_0002.MP4	Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b		
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f		

DJI_0003.MP4	Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a		
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beaacb3		
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192		
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	2	4	2
Flight Logs	33	0	0
Flight Records	2	2	2
Images	6	6	0
Videos	4	4	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	10	10	10
Videos	10	10	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-15 15:01:41 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_external.E01

2026-06-15 15:01:41 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_external.E01

2026-06-15 15:01:41 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_internal.001

2026-06-15 15:01:42 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_internal.001

2026-06-15 15:02:41 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df048-jun-android\p2_image_parsing\controller_android_parsed

out: controller_android_parsed

2026-06-15 15:02:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357829

out: DJI_0001.MP4

2026-06-15 15:02:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357830

out: DJI_0002.MP4

2026-06-15 15:03:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357831

out: DJI_0003.MP4

2026-06-15 15:03:26 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357832

out: DJI_0004.MP4

2026-06-15 15:03:36 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357833

out: DJI_0005.MP4

2026-06-15 15:03:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358853

out: DJI_0001.THM

2026-06-15 15:03:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358854

out: DJI_0002.THM

2026-06-15 15:03:43 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358855

out: DJI_0003.THM

2026-06-15 15:03:43 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358856

out: DJI_0004.THM

2026-06-15 15:03:43 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358857

out: DJI_0005.THM

2026-06-15 15:03:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357829 *out: DJI_0001.MP4*

2026-06-15 15:03:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357830

out: DJI_0002.MP4

2026-06-15 15:04:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357831

out: DJI_0003.MP4

2026-06-15 15:04:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357832

out: DJI_0004.MP4

2026-06-15 15:04:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357833

out: DJI_0005.MP4

2026-06-15 15:04:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358853

out: DJI_0001.THM

2026-06-15 15:04:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358854

out: DJI_0002.THM

2026-06-15 15:04:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358855

out: DJI_0003.THM

2026-06-15 15:04:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358856

out: DJI_0004.THM

2026-06-15 15:04:32 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358857

out: DJI_0005.THM

2026-06-15 15:05:09 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 18-06-19-02-47-38_FLY057.DAT

out: 18-06-19-02-47-38_FLY057.csv, 18-06-19-02-47-38_FLY057.kml

2026-06-15 15:05:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 18-06-19-02-59-05_FLY058.DAT

out: 18-06-19-02-59-05_FLY058.csv, 18-06-19-02-59-05_FLY058.kml

2026-06-15 15:05:14 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19_[14-50-34].txt DJIFlightRecord_2018-06-19_[14-50-34].csv

out: DJIFlightRecord_2018-06-19_[14-50-34].csv

2026-06-15 15:05:15 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-10-10_[10-02-22].txt DJIFlightRecord_2017-10-10_[10-02-22].csv

out: DJIFlightRecord_2017-10-10_[10-02-22].csv

2026-06-15 15:05:15 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_33177fb665068406_1523611394000.jpg

out: thumb_33177fb665068406_1523611394000_exif.json

2026-06-15 15:05:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_5be64acbef28816c_1523610990000.jpg

out: thumb_5be64acbef28816c_1523610990000_exif.json

2026-06-15 15:05:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_6d9f39f0ca9decc3_1523610988000.jpg *out: thumb_6d9f39f0ca9decc3_1523610988000_exif.json*

2026-06-15 15:05:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_7f92b744bbc53b3a_1523610992000.jpg

out: thumb_7f92b744bbc53b3a_1523610992000_exif.json

2026-06-15 15:05:16 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_dd609e61934788b3_1523611494000.jpg

out: thumb_dd609e61934788b3_1523611494000_exif.json

2026-06-15 15:05:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_ed8c1a297e1061b4_1524152656000.jpg

out: thumb_ed8c1a297e1061b4_1524152656000_exif.json

2026-06-15 15:05:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_14_50_39.mp4

out: 2018_06_19_14_50_39_exif.json

2026-06-15 15:05:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_10_10_10_02_47.mp4

out: 2017_10_10_10_02_47_exif.json

2026-06-15 15:05:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-15 15:05:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM

out: DJI_0002_exif.json

2026-06-15 15:05:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-15 15:05:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM

out: DJI_0004_exif.json

2026-06-15 15:05:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM

out: DJI_0005_exif.json

2026-06-15 15:05:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-15 15:05:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4

out: DJI_0002_exif.json

2026-06-15 15:05:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4

out: DJI_0003_exif.json

2026-06-15 15:05:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-15 15:05:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

2026-06-15 15:05:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM *out: DJI_0001_exif.json*

2026-06-15 15:05:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM

out: DJI_0002_exif.json

2026-06-15 15:05:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM

out: DJI_0003_exif.json

2026-06-15 15:05:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM

out: DJI_0004_exif.json

2026-06-15 15:05:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM

out: DJI_0005_exif.json

2026-06-15 15:05:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-15 15:05:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4

out: DJI_0002_exif.json

2026-06-15 15:05:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4

out: DJI_0003_exif.json

2026-06-15 15:05:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-15 15:05:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.